CONSTRUCTIONS AND FINITE VERIFICATIONS FOR STRICT DISMANTLING LEVELS

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ABSTRACT. The recursive dismantling hierarchy measures how a graph can be reduced to one vertex by testing the graphs induced by successive neighbourhoods. This supplement supplies complete seed data and finite proofs for families of strict level k with $4k+4$ and $6k+6$ vertices. We relate constructions with two auxiliary vertices to the graph obtained by deleting edges between the exclusive parts of their profiles, and identify the one-edge completion that produces the eleven-vertex seed of the $3k+8$ family. Finally, we show that adding a layer shifts the reduced homology of every non-central neighbourhood by one degree. The accompanying calculations verify the finite hypotheses, clique counts and homological identities used in these arguments. We also explain and reproduce the finite verification of the least order at strict level two, including its reduction to seven eight-vertex seeds.

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CONTENTS

1. The constructions and their finite data 2
2. The dismantling hierarchy 3
3. The common layer construction 4
3.1. Seed data and adjacency rules 4
3.2. Clique counts and the Euler criterion 5
4. The family of order $4k+4$ 6
4.1. The sixteen-vertex seed and its profiles 6
4.2. The eight- and twelve-vertex prefixes 7
4.3. Certifying the prefixes and the seed 7
4.4. The nineteen neighbourhood obstructions 9
4.5. The family at every level 10
5. The family of order $6k+6$ 10
5.1. The seed and the resulting family 10
5.2. The finite reductions and obstructions 12
6. Completing kernels with two auxiliary types 15

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6.1. Homology for the comparison 15
6.2. The extension and its kernel 17
6.3. The two auxiliary stars 17
6.4. Original and auxiliary neighbourhoods 18
6.5. Equivalent finite conditions and their Euler form 19
6.6. Completing one edge and reducing the remainder 20
6.7. The completion underlying the eleven-vertex seed 21
7. Homology through successive layers 22
7.1. A degree shift in every non-central neighbourhood 22
7.2. The homological criterion 23
7.3. Comparison with Euler characteristic 24
8. The finite verification of the minimum at level two 25
8.1. Why these extensions suffice 25
8.2. Seven seeds with explicit reductions 25
8.3. All choices of the two external neighbourhoods 26
Appendix A. Finite data for the family of order $3k+8$ 27
A.1. Reductions and neighbourhood obstructions 27
A.2. Figure incidences and single deletions 29
References 30

1. The constructions and their finite data

A graph is dismantled by deleting vertices until one vertex remains. In an ordinary dismantling, each deleted vertex is dominated by a neighbour. The recursive hierarchy permits a deletion at level k when the graph induced by its current neighbours is dismantlable at level $k-1$. A graph has *strict level* k if it admits a reduction at level k and none at level $k-1$.

The article *Linear-size graphs at every strict dismantling level* constructs graphs F_k and H_k of strict level k, for every $k \geq 1$, with

$$|V(F_k)| = 3k+8, \qquad |V(H_k)| = 4k+4.$$

The title, authors and current article reference are recorded in `ARTICLE.md`. This supplement develops the finite proof for H_k, gives a third family J_k of order $6k+6$, and explains the kernel completion and homological identities behind the layer construction. Let m_k be the least order of a graph of strict level k. The two families give $m_k \leq \min\{4k+4, 3k+8\}$; they do not determine m_k in general. References here to sections, figures and results of the article concern the identified 23-page manuscript; `ARTICLE.md` distinguishes it from the version at the public article locator.

Each family starts with a finite seed. A new layer contains a centre whose neighbourhood graph is the previous graph, together with auxiliary vertices that exclude first deletions at the lower level. To apply this construction, we need a reduction of the seed at its stated level, an obstruction to the preceding level, a reduction after deleting the first centre, and obstructions in the remaining

neighbourhoods. We supply the finite data for these tasks and use the homological argument of Section 7 to justify the criterion for every subsequent layer.

For example, H_1 has eight vertices. Its reduction at level one requires only two outer deletions, each justified by a single fold in a neighbourhood; the remaining graph is a cone. None of its original neighbourhoods is a cone, so no level-zero reduction can begin. Section 4 gives this example in full before treating the larger recursive certificates.

Organisation. Sections 2 and 3 fix the hierarchy and recall the common construction and its Euler criterion. Section 4 supplies the $4k+4$ family, including its two prefixes; Section 5 checks a second application with five auxiliary types. Section 6 relates constructions with two auxiliary types to an edge-deleted kernel and identifies the completion producing F_1. Its first subsection supplies the homological preliminaries for Sections 7 and 8. Section 7 proves the degree shift in non-central neighbourhoods and the resulting homological criterion. Section 8 explains the finite verification that the minimum at strict level two is twelve. Appendix A records the finite calculations and figures for the article's $3k+8$ family.

Conventions and notation. The layer indices form $\mathcal{L}_m$ (written L_m in the article); clique counts are written κ_t, reserving c_i for centres. The symbols F_k and H_k retain their meanings in the article; J_k denotes the $6k+6$ family. Within a section, A denotes its seed, B its first extension and G_m the graph with m added layers; this layer count is distinct from the minimum order m_k. The seed vertex a_i has numerical label i in the data files. Homology is reduced homology over $\mathbb{F}_2$, with augmented chains in degree -1. The command following each finite calculation runs from the root of the accompanying repository and requires Python 3.11 or later.

2. The dismantling hierarchy

All graphs are finite, simple and undirected. For $S \subseteq V(H)$, write $H[S]$ for the induced subgraph and $H - v = H[V(H) \setminus \{v\}]$. The open and closed neighbourhood sets are

$$N_H(v) = \{u : uv \in E(H)\}, \qquad N_H[v] = N_H(v) \cup \{v\}.$$

We write $L_H(v) = H[N_H(v)]$ for the *neighbourhood graph*. A *cone* is a non-empty graph with a universal vertex, called an *apex*. Thus K_1 is a cone and the empty graph is not.

A vertex u is *dominated* by $w \neq u$ when $N_H[u] \subseteq N_H[w]$. Deleting it is a *fold*, written $u \to w$. All neighbourhoods in a sequence of folds are taken in the graph then remaining.

Lemma 2.1. *A vertex u is dominated if and only if $L_H(u)$ is a cone.*

Proof. The inclusion $N_H[u] \subseteq N_H[w]$ says exactly that w is a neighbour of u adjacent to all its other neighbours. □

The hierarchy of Fieux and Jouve [3] replaces the cone test by a reduction in the preceding class. At level one these are the s-dismantling moves of [1].

Definition 2.2. *Let $\mathcal{D}_{-1}$ consist of the non-empty cones. For $j \geq 0$, a graph is in $\mathcal{D}_j$ if it reduces to K_1 by successive deletions whose current neighbourhood graphs belong to $\mathcal{D}_{j-1}$. Such a graph has strict level j when it is outside $\mathcal{D}_{j-1}$. The empty graph belongs to none of these classes.*

The empty reduction puts K_1 in every class. Membership is invariant under isomorphism, since an isomorphism preserves every neighbourhood test.

Lemma 2.3. *The classes are nested:* $\mathcal{D}_{-1} \subseteq \mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \cdots$.

Proof. A cone folds onto its apex. If $\mathcal{D}_{j-1} \subseteq \mathcal{D}_j$, every neighbourhood in a level-j reduction also passes the level-$(j+1)$ test. The same sequence proves the next inclusion. □

Lemma 2.4. *For* $|V(H)| > 1$ *and* $j \geq 0$,

$$H \in \mathcal{D}_j \iff \text{some } v \in V(H) \text{ satisfies } L_H(v) \in \mathcal{D}_{j-1} \text{ and } H - v \in \mathcal{D}_j. \tag{1}$$

Proof. Read the first deletion of a reduction for the forward implication. For the converse, prepend the permitted deletion of v to a reduction of $H - v$. □

In particular, failure of the neighbourhood test at every vertex excludes membership. Equation (1), together with the empty graph and cone cases, is also the finite recursion used by the verification programs.

3. The common layer construction

The three families use the same adjacency rules. We recall them here so that the seed data in the next sections determine each graph without consulting another document. A centre will recover the preceding graph as its neighbourhood; auxiliary vertices will provide the obstructions that make the next level strict.

3.1. **Seed data and adjacency rules.** Choose a non-empty seed A, an integer $d \geq 1$, and a graph R on the type set $\mathcal{T} = \{0, \ldots, d-1\}$. For each type j, choose a *profile* $P_j \subseteq V(A)$: these are its neighbours in the seed. Let $\mathcal{L}_m = \{0, \ldots, m-1\}$, with $\mathcal{L}_0 = \varnothing$. The graph G_m has vertex set

$$V(A) \sqcup \{c_i : i \in \mathcal{L}_m\} \sqcup \{x_{j,i} : j \in \mathcal{T},\ i \in \mathcal{L}_m\}.$$

Here c_i is the centre of layer i and $x_{j,i}$ is its auxiliary of type j. The following rules specify all edges.

(1) The seed induces A.
(2) The centres form a clique and are adjacent to every seed vertex.
(3) All copies of a fixed auxiliary type form a clique.
(4) Auxiliaries of distinct types j, t are adjacent exactly when $jt \in E(R)$.
(5) A seed vertex a is adjacent to $x_{j,i}$ exactly when $a \in P_j$.
(6) The centre c_i is adjacent to $x_{j,h}$ exactly when $i \neq h$.

Put $B = G_1$, $p = c_0$ and $x_j = x_{j,0}$. Then $G_0 = A$, $L_B(p) = A$, and

$$|V(G_m)| = |V(A)| + (d+1)m. \tag{2}$$

A vertex other than a centre is called *non-central*.

The following two facts supply the positive reductions and will also be used in the homological arguments.

Lemma 3.1. *For* $m \geq 1$ *and* $i \in \mathcal{L}_m$, $L_{G_m}(c_i) \cong G_{m-1}$.

Proof. Its vertices are the seed and all layers except layer i. Fix the seed and renumber the remaining layers in increasing order. All six rules are preserved, including the inequality of layer indices in the last rule. □

Lemma 3.2. *For* $m \geq 2$, *the graph* $G_m - c_i$ *belongs to* $\mathcal{D}_0$.

Proof. For each type j and $h \neq i$, fold $x_{j,h}$ onto $x_{j,i}$. These vertices have the same neighbours in the seed and among auxiliary types; after c_i is deleted, the target sees all remaining centres and the source omits c_h. Hence the required closed-neighbourhood inclusion holds. The inclusions persist after preceding folds, and every target is retained. After all folds, each remaining centre is universal. At least one exists because $m \geq 2$, so the resulting cone folds to a vertex. □

For $m = 1$ there is no remaining centre. The reduction of $B - p$ must be checked separately for each seed.

3.2. **Clique counts and the Euler criterion.** Let $\kappa_t(H)$ count cliques of cardinality t, with $\kappa_0(H) = 1$. The *clique polynomial* and its alternating sum are

$$C_H(z) = \sum_{t \geq 0} \kappa_t(H) z^t, \qquad q(H) = C_H(-1).$$

The *clique complex* $\mathrm{Cl}(H)$ consists of the cliques, including the empty face. With reduced Euler characteristic, including the empty graph convention,

$$q(H) = 1 - \chi(\mathrm{Cl}(H)) = -\widetilde{\chi}(\mathrm{Cl}(H)), \qquad q(\varnothing) = 1. \tag{3}$$

Lemma 3.3. *For every vertex v,*

$$C_H(z) = C_{H-v}(z) + z C_{L_H(v)}(z), \qquad q(H) = q(H - v) - q(L_H(v)).$$

A cone has $q = 0$, and a fold preserves q.

Proof. Separate cliques according to whether they contain v [4]. In a cone, adjoining the apex pairs cliques of opposite signs. For a fold, Lemma 2.1 makes the neighbourhood term vanish. □

Taking coefficients gives

$$\kappa_t(H) = \kappa_t(H - v) + \kappa_{t-1}(L_H(v)) \quad (t \geq 1). \tag{4}$$

The initial values are $\kappa_0(H) = 1$ and $\kappa_t(\varnothing) = 0$ for $t \geq 1$.

Lemma 3.4. *Membership in any $\mathcal{D}_j$, $j \geq -1$, implies $q = 0$.*

Proof. Start with cones. Inductively, all neighbourhood terms in a level-j reduction vanish. The recurrence preserves q down to K_1, where it is zero. □

Thus a non-zero q excludes membership; a zero value does not establish it.

Theorem 3.5. *Suppose that, for some $s \geq 0$,*

1. $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$;
2. $B - p \in \mathcal{D}_s$;
3. $q(L_B(v)) \neq 0$ *for every* $v \neq p$.

Then $G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1}$ for every $m \geq 0$, with order given by (2).

This is the Euler criterion used in the article. A proof within this supplement follows from Theorem 7.2: by (19), a non-zero q gives non-zero reduced homology, and $\mathcal{D}_s \subseteq \mathcal{D}_{s+1}$ supplies its remainder hypothesis. Section 7 proves that theorem from the adjacency rules, Lemmas 3.1 and 3.2, the first-deletion equivalence (1), and the elementary homology of Section 6.1. No other result of Sections 4–6 is used.

Remark 3.6. *The same argument allows* $B - p \in \mathcal{D}_{s+1}$ *in place of condition (2). It does not remove the requirement to reduce* $B - p$.

The article also proves that these Euler hypotheses require at least two auxiliary types, hence at least three new vertices per layer. This restriction concerns the stated template. It is attained by F_k and is not a lower bound on the order of every graph of strict level k.

4. The family of order $4k + 4$

The second family in the article begins with graphs on eight, twelve and sixteen vertices at strict levels one, two and three. All three occur as nested induced subgraphs. From the sixteen-vertex seed onwards, the layer construction adds four vertices at a time. We specify these graphs, justify their finite reductions and obstructions, and then apply the general theorem.

4.1. **The sixteen-vertex seed and its profiles.** We use three auxiliary types, so each layer adds four vertices. Let $V(A) = \{a_0, \dots, a_{15}\}$, with the adjacency list in Table 1. It is symmetric, has no loops, and has 97 edges. The profiles in Table 2 have sizes 13, 14 and 15. The type graph is

$$E(R) = \{\{0, 1\}, \{0, 2\}\}. \tag{5}$$

TABLE 1. Complete adjacency list of the sixteen-vertex seed A.

v	$N_A(a_v)$, given by indices
0	2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14
1	2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14
2	0, 1, 3, 6, 7, 8, 9, 10, 11, 12, 13, 15
3	0, 1, 2, 4, 5, 8, 9, 10, 11, 12, 13, 15
4	1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15
5	0, 3, 4, 6, 7, 9, 10, 11, 12, 13, 14, 15
6	0, 1, 2, 4, 5, 8, 9, 11, 12, 13, 14, 15
7	0, 1, 2, 4, 5, 8, 9, 10, 12, 13, 14, 15
8	0, 1, 2, 3, 4, 6, 7, 9, 10, 12, 14, 15
9	0, 1, 2, 3, 5, 6, 7, 8, 11, 13, 14, 15
10	0, 1, 2, 3, 4, 5, 7, 8, 11, 13, 14, 15
11	0, 1, 2, 3, 4, 5, 6, 9, 10, 12, 14, 15
12	0, 1, 2, 3, 4, 5, 6, 7, 8, 11, 14, 15
13	0, 1, 2, 3, 4, 5, 6, 7, 9, 10, 14, 15
14	0, 1, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15
15	2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14

TABLE 2. Profiles specified by their omitted seed vertices. Every copy of type j is adjacent to all remaining vertices of A.

j	$V(A) \setminus P_j$	$\lvert P_j \rvert$
0	$\{a_8, a_{11}, a_{14}\}$	13
1	$\{a_{11}, a_{15}\}$	14
2	$\{a_{13}\}$	15

The graph $B = G_1$ adds $p = c_0$ and x_0, x_1, x_2 to A. The centre is adjacent to every vertex of A and to no auxiliary. The neighbours of x_j in A are P_j, and its auxiliary adjacencies are specified by R. Thus

$$|V(B)| = 20, \qquad |E(B)| = 97 + 16 + 42 + 2 = 157.$$

For the finite certificates, a_i has index i, $p = 16$, and $x_j = 17 + j$.

4.2. **The eight- and twelve-vertex prefixes.** The first two members are induced subgraphs of the same seed. Put

$$S_1 = \{2, 4, 5, 8, 9, 10, 12, 13\},$$
$$S_2 = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13\},$$

and define

$$H_1 = A[\{a_i : i \in S_1\}], \qquad H_2 = A[\{a_i : i \in S_2\}], \qquad H_3 = A.$$

The direct reduction below establishes the strict level of the first prefix. The recursive certificates then establish the strict level of the second prefix and of the sixteen-vertex seed. The two prefixes have 8 and 12 vertices, respectively. The inclusions $S_1 \subset S_2 \subset V(A)$ give induced inclusions. We now verify the first prefix directly. Use numerical labels $a_i = i$. Restricting the adjacency table of A gives

v	$N_{H_1}(v)$
2	8, 9, 10, 12, 13
4	5, 8, 10, 12, 13
5	4, 9, 10, 12, 13
8	2, 4, 9, 10, 12
9	2, 5, 8, 13
10	2, 4, 5, 8, 13
12	2, 4, 5, 8
13	2, 4, 5, 9, 10

Proposition 4.1. *The graph H_1 belongs to $\mathcal{D}_1 \setminus \mathcal{D}_0$.*

Proof. First consider $H_1[N_{H_1}(5)]$, on $\{4, 9, 10, 12, 13\}$. The only neighbour of 9 there is 13, so $9 \to 13$ is a fold. The remaining graph on $\{4, 10, 12, 13\}$ is a cone with apex 4. Consequently 5 can be deleted from H_1 at level one.

In $H_1 - 5$, the neighbourhood of 4 is $\{8, 10, 12, 13\}$. Within this neighbourhood, $12 \to 8$ is a fold, and the remaining path $8, 10, 13$ is a cone with apex 10. Thus 4 can next be deleted at level one. The graph $H_1 - \{5, 4\}$ is a cone with apex 2, proving $H_1 \in \mathcal{D}_1$. The inner folds justify only the stated outer deletion; they do not delete vertices from the outer graph.

To exclude level zero, Table 3 lists, for every v, each candidate apex $t \in N_{H_1}(v)$ and a vertex $w \in N_{H_1}(v) \setminus \{t\}$ not adjacent to it. Every candidate occurs, so no neighbourhood has an apex. There is no first fold, and $H_1 \notin \mathcal{D}_0$. □

4.3. **Certifying the prefixes and the seed.** For the larger graphs, the neighbourhood reductions are themselves recursive. We record them as finite proof states so that every required membership and non-membership can be checked. The main certificate contains 1,114 proof nodes for A and 184 for $B - p$. The prefix certificates use the ordered vertex lists S_1, S_2 of this section. All neighbourhoods and domination relations are computed in the current induced subgraph.

TABLE 3. Complete exclusion of an apex in every neighbourhood of H_1.

v	Candidate apex $t \mapsto w$, with $tw \notin E(H_1)$
2	$8 \mapsto 13,\ 9 \mapsto 10,\ 10 \mapsto 9,\ 12 \mapsto 9,\ 13 \mapsto 8$
4	$5 \mapsto 8,\ 8 \mapsto 5,\ 10 \mapsto 12,\ 12 \mapsto 10,\ 13 \mapsto 8$
5	$4 \mapsto 9,\ 9 \mapsto 4,\ 10 \mapsto 9,\ 12 \mapsto 9,\ 13 \mapsto 12$
8	$2 \mapsto 4,\ 4 \mapsto 2,\ 9 \mapsto 4,\ 10 \mapsto 9,\ 12 \mapsto 9$
9	$2 \mapsto 5,\ 5 \mapsto 2,\ 8 \mapsto 5,\ 13 \mapsto 8$
10	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 8 \mapsto 5,\ 13 \mapsto 8$
12	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 8 \mapsto 5$
13	$2 \mapsto 4,\ 4 \mapsto 2,\ 5 \mapsto 2,\ 9 \mapsto 4,\ 10 \mapsto 9$

4.3.1. *The recursive proof rule.* A proof state (S, j), with $j \geq -1$, asserts whether the graph induced by S belongs to $\mathcal{D}_j$. There are five allowed forms of node.

(1) The empty state is false at every level.
(2) A non-empty cone is true at every level, with its apex supplied.
(3) A non-empty non-cone is false at level -1.
(4) A true state with $j \geq 0$ specifies $v \in S$, a true child $(S \cap N(v), j-1)$, and a true child $(S \setminus \{v\}, j)$.
(5) A non-empty non-cone state (S, j), with $j \geq 0$ (and hence $|S| \geq 2$), is false if it specifies, for every $v \in S$, a false child of at least one of those two forms.

The checker verifies the exact subsets, levels, truth values and child references, requires every vertex in a negative node to be covered, and rejects cycles. Every child has fewer vertices. It reconstructs each induced graph from the stated adjacency data, tests every proposed apex, and checks the absence of an apex in each negative non-cone state. A deleted vertex must belong to the current set. The correctness of a checked proof graph therefore follows by induction on the number of vertices, using Definition 2.2 and Lemma 2.3 for cone states. By the first-deletion equivalence (1), any reduction of a state with $|S| \geq 2$ begins with a vertex v such that $(S \cap N(v), j-1)$ is true, and the remaining reduction shows that $(S \setminus \{v\}, j)$ is true. The false children of a form-5 node exclude every such v. This gives a finite certificate of non-membership as well as membership, without relying on a bounded unsuccessful search.

The three main roots are bound to

$$(V(A), 3, \text{true}), \quad (V(A), 2, \text{false}), \quad (V(B) \setminus \{p\}, 3, \text{true}).$$

The four prefix roots assert membership at levels $1, 2$ and non-membership at levels $0, 1$, respectively. The checker also verifies the prefix adjacencies against the specified induced subgraphs of A. The seed and first extension are reconstructed from Tables 1 and 2, the type graph in (5), and the six adjacency rules of Section 3.

The complete finite proof graphs are the files `data/certificates/FINITE_CERTIFICATE.json.gz` and `data/certificates/PREFIX_CERTIFICATES.json.gz`. The seed and profiles are in `data/certificates/PARAMETERS.json`; the ordered prefix sets and their adjacencies are in `data/certificates/PREFIX.json`. The checking rules are implemented in `dismantling/certificates.py`. The verifier checks the supplied proof states using integer and set operations; it does not search for reductions.

4.3.2. *The reductions and their obstructions.* For H_1, delete a_5, a_4 in that order; the remaining graph is a cone with apex a_2. Their current neighbourhoods belong to $\mathcal{D}_0$, as proved by the referenced child certificates. For H_2, delete a_6, a_5, a_4; the current neighbourhoods belong to $\mathcal{D}_1$, and the remainder is a cone with apex a_2. For $A = H_3$, delete a_0, a_1; the current neighbourhoods belong to $\mathcal{D}_2$, and the remainder is a cone with apex a_{15}. For $B - p$, the outer order is

$$a_3, a_1, a_2, a_0, a_4, a_6, a_5, a_8, a_9, a_{10}, a_{11}.$$

Each current neighbourhood belongs to $\mathcal{D}_2$, and the remainder is a cone with apex a_7. These orders alone do not certify their neighbourhoods; the complete child proof graphs do so by the preceding rules.

The negative root for $A \notin \mathcal{D}_2$ has one obstruction for each of its 16 vertices: every $A[N_A(a_i)]$ is certified outside $\mathcal{D}_1$. The prefix negative roots similarly provide a complete obstruction for every possible first deletion at their claimed lower levels. A zero value of q is never used as a sufficient test of membership.

4.4. **The nineteen neighbourhood obstructions.** Table 4 supplies all coefficients through degree seven of every non-central neighbourhood in B. All higher degrees were computed and verified to vanish. They are checked using the coefficient recurrence (4). The supplied checker uses this deletion recurrence to verify every coefficient and alternating sum in the distributed certificate. For example, the row for a_0 gives $1 - 16 + 95 - 263 + 344 - 182 + 20 = -1$. All 19 sums are non-zero.

TABLE 4. Clique coefficients of $B[N_B(v)]$, for every non-central vertex. All coefficients of degree greater than seven are zero. The last column is their alternating sum.

v	κ_0	κ_1	κ_2	κ_3	κ_4	κ_5	κ_6	κ_7	q
a_0	1	16	95	263	344	182	20	0	−1
a_1	1	16	95	262	340	179	20	0	−1
a_2	1	16	96	271	362	189	10	0	−7
a_3	1	16	95	261	335	172	16	0	−2
a_4	1	16	96	275	391	257	62	3	−1
a_5	1	16	97	281	402	262	60	3	−2
a_6	1	16	95	263	349	200	34	2	−2
a_7	1	16	97	279	388	232	39	1	−3
a_8	1	15	86	235	308	164	15	0	−4
a_9	1	16	95	263	344	183	19	0	−3
a_{10}	1	16	95	264	354	207	37	1	−1
a_{11}	1	14	75	194	249	143	27	2	−1
a_{12}	1	16	96	271	367	211	34	2	−2
a_{13}	1	15	86	237	323	199	41	1	−1
a_{14}	1	16	97	281	402	262	60	3	−2
a_{15}	1	16	96	272	375	230	48	3	−1
x_0	1	15	86	236	315	179	26	0	−2
x_1	1	15	85	228	296	167	27	0	−1
x_2	1	16	97	278	380	213	27	1	−3

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only 4k4
```

The output reports seven checked roots and nineteen clique rows. The root checks certify $H_1 \in \mathcal{D}_1 \setminus \mathcal{D}_0$, $H_2 \in \mathcal{D}_2 \setminus \mathcal{D}_1$, $A \in \mathcal{D}_3 \setminus \mathcal{D}_2$, and $B - p \in \mathcal{D}_3$. Every coefficient in the printed table is compared with both clique enumeration and the deletion recurrence, including the vanishing of higher coefficients.

4.5. **The family at every level.** The certificates establish the hypotheses needed for the extension from the sixteen-vertex seed. Theorem 3.5 then treats any number of layers.

Theorem 4.2. *With the seed and profiles in this section, put* $H_3 = A$ *and* $H_k = G_{k-3}$ *for* $k \geq 3$*, and use the two induced prefixes for* $k = 1, 2$*. Then*

$$H_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}, \qquad |V(H_k)| = 4k + 4 \qquad (k \geq 1).$$

Every inclusion $H_k \subset H_{k+1}$ *is induced.*

Proof. The seven roots give the two prefixes and establish $A \in \mathcal{D}_3 \setminus \mathcal{D}_2$ and $B - p \in \mathcal{D}_3$. The nineteen non-zero sums in Table 4 exclude every non-central vertex as a first deletion. Thus Theorem 3.5 applies with $s = 3$ and $d = 3$. For $k \geq 3$ it gives strict level k and $16 + 4(k - 3) = 4k + 4$ vertices. The prefix certificates prove the cases $k = 1, 2$. The nested prefix sets and the layer rules give the induced inclusions. □

For $k \geq 3$, put $m = k - 3$. The layered graphs also have

$$|E(G_m)| = 97 + 16m + 42m + 4\binom{m}{2} + 2m^2 + 3m(m-1) = 97 + 53m + 7m^2.$$

The terms count seed edges, seed–centre edges, seed–auxiliary edges, edges between centres or within one type, edges between the two adjacent pairs of types, and centre–auxiliary edges, respectively.

5. The family of order $6k + 6$

The same layer construction also applies to a twelve-vertex seed at strict level one. This example uses five auxiliary types, and its positive certificates can be written out entirely as folds inside successive neighbourhoods. It illustrates how the three finite hypotheses are checked when no recursive certificate file is needed. We first state the finite assertions and their consequences; the reductions and clique tables used in the proof are displayed in Section 5.2.

5.1. **The seed and the resulting family.** Here five auxiliary types give six new vertices per layer. Let $V(A) = \{a_0, \ldots, a_{11}\}$, with the adjacency list in Table 5. The list is symmetric, has no loops, and gives $|E(A)| = 47$. The profiles are given in Table 6; their cardinalities sum to 48. The type graph is

$$E(R) = \{02, 03, 04, 12, 14, 23, 34\}, \tag{6}$$

where jt abbreviates the unordered pair $\{j, t\}$. There are no other edges of R.

TABLE 5. The complete adjacency list of A. An integer i denotes a_i.

v	$N_A(v)$
0	$\{1,3,4,5,6,7,9,10\}$
1	$\{0,2,3,4,5,6,7,8,10\}$
2	$\{1,4,5,6,8,9,11\}$
3	$\{0,1,4,6,7,8,9,11\}$
4	$\{0,1,2,3,5,8,10,11\}$
5	$\{0,1,2,4,6,9,10,11\}$
6	$\{0,1,2,3,5,8,9\}$
7	$\{0,1,3,9,10,11\}$
8	$\{1,2,3,4,6,9,10,11\}$
9	$\{0,2,3,5,6,7,8,10,11\}$
10	$\{0,1,4,5,7,8,9,11\}$
11	$\{2,3,4,5,7,8,9,10\}$

TABLE 6. The five profiles. Each displayed integer i denotes a_i.

j	Indices of the vertices in P_j
0	1, 2, 3, 4, 5, 6, 7, 8, 9, 11
1	1, 3, 4, 6, 7, 8, 10, 11
2	0, 1, 3, 4, 5, 6, 7, 9, 10, 11
3	1, 2, 3, 5, 6, 7, 8, 9, 10, 11
4	0, 1, 2, 3, 5, 7, 8, 9, 10, 11

The graph B consists of A, the centre p, and the five auxiliaries $x_0, \dots, x_4$. Its centre is adjacent to every vertex of A and to no auxiliary; the neighbours of x_j in A are precisely P_j, and its auxiliary adjacencies are those of R. Thus

$$|V(B)| = 18, \qquad |E(B)| = 47 + 12 + 48 + 7 = 114.$$

In the finite certificates, a_i is abbreviated to i, and $p = 12$, $x_j = 13 + j$. In particular, vertex 15 denotes x_2.

Proposition 5.1. *These seed data satisfy*

$$A \in \mathcal{D}_1 \setminus \mathcal{D}_0, \qquad B - p \in \mathcal{D}_1, \qquad q(B[N_B(v)]) = \begin{cases} 2, & v = a_7 \text{ or } v = x_2, \\ 1, & v \neq p, a_7, x_2 \end{cases} \quad (v \neq p).$$

Proof. Table 7 gives, for every $v \in V(A)$, the degree $d_v = |N_A(v)|$ and the maximum degree in $A[N_A(v)]$. In each row that maximum is strictly smaller than $d_v - 1$. No open neighbourhood graph is a cone; Lemma 2.4 therefore gives $A \notin \mathcal{D}_0$.

Tables 8 and 9 specify successive outer deletions in A and $B - p$, respectively. Each row lists the exact neighbourhood set in the current outer graph, the folds within that neighbourhood in the stated order, and an apex of the resulting neighbourhood. All fourteen required closed neighbourhood inclusions are displayed in Table 10. They are obtained by restricting the adjacency data of Tables 5 and 6 and (6) to the current set. After the outer deletions, the remaining vertex

sets are cones with the apices specified in Section 5.2. Each inner reduction is consequently a level-0 reduction, and each outer deletion is permitted at level 1. The final cones also lie in $\mathcal{D}_1$, by Lemma 2.3. This proves $A \in \mathcal{D}_1$ and $B - p \in \mathcal{D}_1$.

Table 11 gives all clique coefficients of the seventeen non-central neighbourhood graphs. These coefficients follow by the coefficient recurrence in Lemma 3.3, with $\kappa_0(H) = 1$ and $\kappa_t(\varnothing) = 0$ for $t \geq 1$. Their alternating sums are 2 in rows 7 and 15, and 1 in every other row. This proves the last assertion. □

For this seed, the graphs of Section 3 satisfy

$$|V(G_m)| = 12 + 6m, \qquad |E(G_m)| = 47 + 52m + 15m^2. \tag{7}$$

Indeed, the seven disjoint classes of edges give

$$\begin{aligned} |E(G_m)| &= 47 + 12m + 48m + \binom{m}{2} + 5\binom{m}{2} + 7m^2 + 5m(m-1) \\ &= 47 + 52m + 15m^2. \end{aligned}$$

The terms count, in order, edges within A, between A and centres, between A and auxiliaries, between centres, within each auxiliary type, between distinct types, and between centres and auxiliaries.

Theorem 5.2. *For every $k \geq 1$, put $J_k = G_{k-1}$. Then $J_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}$ and $|V(J_k)| = 6k + 6$.*

Proof. Proposition 5.1 supplies the three hypotheses of Theorem 3.5, with $s = 1$ and $d = 5$. Consequently

$$G_m \in \mathcal{D}_{m+1} \setminus \mathcal{D}_m \qquad (m \geq 0).$$

Taking $m = k - 1$ gives $J_k \in \mathcal{D}_k \setminus \mathcal{D}_{k-1}$ and $|V(J_k)| = 12 + 6(k-1) = 6k + 6$. □

5.2. **The finite reductions and obstructions.** The certificates below use only the adjacency list, the profiles, and the type edges specified in Section 5.1.

5.2.1. *Excluding a first fold in A.* For $v \in V(A)$, put $d_v = |N_A(v)|$, and write $\Delta(Q)$ for the maximum degree in a non-empty graph Q. The entries of Table 7 satisfy $\Delta(A[N_A(v)]) < d_v - 1$, so none of these neighbourhoods has an apex.

TABLE 7. Every open neighbourhood graph of A has no universal vertex.

v	$d_v = \lvert N_A(v)\rvert$	$d_v - 1$	$\Delta(A[N_A(v)])$
0	8	7	6
1	9	8	6
2	7	6	5
3	8	7	5
4	8	7	6
5	8	7	5
6	7	6	5
7	6	5	4
8	8	7	5
9	9	8	6
10	8	7	5
11	8	7	6

5.2.2. *The two level-1 reductions.* In each row of Tables 8 and 9, v is deleted from the outer graph after all preceding outer deletions. The set S is its neighbourhood set at that moment. The arrows are folds inside the graph induced by S. After each arrow its source vertex is removed before the next inclusion is evaluated. The listed apex is universal after those inner folds. A row with no folds already has a conical neighbourhood graph.

For Table 8, the final outer vertex set is $\{5, 6, 8, 9, 10, 11\}$, with apex 9. For Table 9, it is $\{3, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17\}$, with apex 11. The centre p has been removed before Table 9 begins.

TABLE 8. A reduction certificate for $A \in \mathcal{D}_1$.

v	Current neighbourhood set S	Folds in order	Apex
7	$\{0, 1, 3, 9, 10, 11\}$	$1 \to 0$	9
0	$\{1, 3, 4, 5, 6, 9, 10\}$	$4 \to 1$, $3 \to 6$	5
1	$\{2, 3, 4, 5, 6, 8, 10\}$	$3 \to 8$, $6 \to 2$	4
2	$\{4, 5, 6, 8, 9, 11\}$	$4 \to 11$	9
3	$\{4, 6, 8, 9, 11\}$	none	8
4	$\{5, 8, 10, 11\}$	none	10

TABLE 9. A reduction certificate for $B - p \in \mathcal{D}_1$.

v	Current neighbourhood set S	Folds in order	Apex
0	$\{1,3,4,5,6,7,9,10,15,17\}$	$4 \to 1$, $6 \to 15$, $3 \to 7$	10
2	$\{1,4,5,6,8,9,11,13,16,17\}$	none	13
4	$\{1,3,5,8,10,11,13,14,15\}$	$5 \to 15$, $10 \to 14$	3
1	$\{3,5,6,7,8,10,13,14,15,16,17\}$	$5 \to 16$, $6 \to 3$, $8 \to 17$	7
5	$\{6,9,10,11,13,15,16,17\}$	none	9
6	$\{3,8,9,13,14,15,16\}$	none	3

For example, the first row of Table 8 starts with $S = \{0,1,3,9,10,11\}$. Within this neighbourhood,

$$N_{A[S]}[1] = \{0,1,3,10\} \subseteq N_{A[S]}[0] = \{0,1,3,9,10\}.$$

The fold $1 \to 0$ leaves a cone with apex 9, so the outer vertex 7 can be deleted at level 1. The next row is evaluated in $A - 7$. Table 10 provides the corresponding sets for every fold in both certificates. Its third column is contained in its fourth in each row; both columns are the closed neighbourhoods in the current inner graph.

TABLE 10. Closed neighbourhood inclusions for all fourteen folds. Here T is the neighbourhood graph after the preceding folds in the same row of the corresponding certificate.

Graph; outer v	Fold $u \to t$	$N_T[u]$	$N_T[t]$
A; 7	$1 \to 0$	$\{0,1,3,10\}$	$\{0,1,3,9,10\}$
A; 0	$4 \to 1$	$\{1,3,4,5,10\}$	$\{1,3,4,5,6,10\}$
A; 0	$3 \to 6$	$\{1,3,6,9\}$	$\{1,3,5,6,9\}$
A; 1	$3 \to 8$	$\{3,4,6,8\}$	$\{2,3,4,6,8,10\}$
A; 1	$6 \to 2$	$\{2,5,6,8\}$	$\{2,4,5,6,8\}$
A; 2	$4 \to 11$	$\{4,5,8,11\}$	$\{4,5,8,9,11\}$
$B - p$; 0	$4 \to 1$	$\{1,3,4,5,10,15\}$	$\{1,3,4,5,6,7,10,15,17\}$
$B - p$; 0	$6 \to 15$	$\{1,3,5,6,9,15\}$	$\{1,3,5,6,7,9,10,15\}$
$B - p$; 0	$3 \to 7$	$\{1,3,7,9,15,17\}$	$\{1,3,7,9,10,15,17\}$
$B - p$; 4	$5 \to 15$	$\{1,5,10,11,13,15\}$	$\{1,3,5,10,11,13,14,15\}$
$B - p$; 4	$10 \to 14$	$\{1,8,10,11,14,15\}$	$\{1,3,8,10,11,14,15\}$
$B - p$; 1	$5 \to 16$	$\{5,6,10,13,15,16,17\}$	$\{3,5,6,7,8,10,13,15,16,17\}$
$B - p$; 1	$6 \to 3$	$\{3,6,8,13,14,15,16\}$	$\{3,6,7,8,13,14,15,16,17\}$
$B - p$; 1	$8 \to 17$	$\{3,8,10,13,14,16,17\}$	$\{3,7,8,10,13,14,16,17\}$

5.2.3. *Clique coefficients.* Table 11 supplies the coefficients $\kappa_0, \ldots, \kappa_6$ of every non-central neighbourhood graph in B. There are no cliques of greater cardinality in these graphs. The coefficient recurrence (4) computes every row from the adjacency data. For instance, the alternating sums in rows 0 and 7 are

$$1 - 11 + 41 - 63 + 35 - 2 = 1, \qquad 1 - 12 + 49 - 86 + 63 - 15 + 2 = 2.$$

All seventeen sums are non-zero.

TABLE 11. Clique counts for the seventeen non-central open neighbourhood graphs of B. Coefficients κ_t with $t > 6$ are zero.

v	κ_0	κ_1	κ_2	κ_3	κ_4	κ_5	κ_6	q
0	1	11	41	63	35	2	0	1
1	1	15	72	133	88	12	0	1
2	1	11	43	75	58	17	2	1
3	1	14	65	122	91	23	3	1
4	1	12	46	71	39	2	0	1
5	1	13	57	103	74	16	1	1
6	1	12	46	75	51	10	0	1
7	1	12	49	86	63	15	2	2
8	1	13	57	103	75	18	2	1
9	1	14	66	131	111	37	5	1
10	1	13	56	97	61	7	0	1
11	1	14	65	125	101	32	5	1
13	1	13	59	119	110	42	5	1
14	1	10	33	45	22	0	0	1
15	1	13	57	103	74	15	1	2
16	1	13	60	125	120	47	5	1
17	1	13	56	104	85	28	4	1

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only 6k6
```

The output confirms twelve outer deletions, fourteen inner folds, twelve rows excluding a first fold in the seed, and seventeen clique rows. Each displayed neighbourhood and both sides of every fold inclusion are recomputed from the adjacency lists. The calculation also verifies the final cones in the two reductions.

6. COMPLETING KERNELS WITH TWO AUXILIARY TYPES

The family F_k adds only two auxiliaries per layer. For this case we can understand the neighbourhood obstructions by removing certain edges from the seed. We first introduce the homology used to compare the resulting graphs, then derive the identities and apply them to the eleven-vertex seed.

6.1. Homology for the comparison.

A non-zero reduced homology group excludes a graph from every dismantling class. We give the chain conventions and the elementary facts used in this section and in Section 7; in particular, empty neighbourhoods must remain visible in degree -1. The clique complex $\mathrm{Cl}(H)$ contains the cliques of H, including the empty face. We use *augmented reduced homology over* $\mathbb{F}_2$. For a complex K, the chain group $\widetilde{C}_t(K)$ has the faces of cardinality $t+1$ as a basis when $t \geq 0$, and $\widetilde{C}_{-1}(K) = \mathbb{F}_2$, generated by the empty face. Lower chain groups are zero. The boundary of a face is the sum of its codimension-one faces; in particular the boundary of a vertex is the empty face. Each term occurs twice in ∂^2, so this is a chain complex.

We write $\widetilde{H}_t(K) = \ker \partial_t / \operatorname{im} \partial_{t+1}$ and call K *acyclic* if every $\widetilde{H}_t(K)$ vanishes. In particular,

$$\widetilde{H}_{-1}(\{\varnothing\}) = \mathbb{F}_2, \qquad \widetilde{H}_t(\{\varnothing\}) = 0 \quad (t \neq -1). \tag{8}$$

A complex with a vertex has zero homology in degree -1. The *reduced Betti number* of a complex K in degree t is $\widetilde{\beta}_t(K) = \dim_{\mathbb{F}_2} \widetilde{H}_t(K)$. Acyclicity is used as a homological condition, not as a synonym for contractibility or for membership in any $\mathcal{D}_j$. When a homological term is applied to a graph, it refers to its clique complex; in particular, *acyclic* here does not mean that the graph is a forest.

A *simplicial cone* with apex w is a complex $w * J$ whose faces are exactly the faces σ of J and the sets $\sigma \cup \{w\}$, where $w \notin V(J)$. In particular, if a graph H is a cone with apex w, then $\mathrm{Cl}(H) = w * \mathrm{Cl}(H - w)$.

Lemma 6.1. *A simplicial cone is acyclic.*

Proof. For an apex w, let h adjoin w to a face not containing it and send a face containing w to zero. Direct expansion gives $\partial h + h\partial = \mathrm{id}$, including on the empty face. Every cycle z is therefore the boundary of $h(z)$. □

We will compare complexes by expressing them as unions of acyclic pieces.

Lemma 6.2. *If $M = M_1 \cup M_2$ and M_1, M_2 are acyclic, then for every integer t,*

$$\widetilde{H}_t(M) \cong \widetilde{H}_{t-1}(M_1 \cap M_2).$$

The intersection may be $\{\varnothing\}$.

Proof. For $t \geq 0$, write a cycle $z = u + v$, with chains in M_1, M_2. Then $\partial u = \partial v = b$ is a cycle in $M_1 \cap M_2$. Changing the decomposition adds the same chain of the intersection to u and v, so it changes b by a boundary. If z is changed by a boundary, decompose a filling into chains of M_1, M_2; the resulting class of b is unchanged. This defines $[z] \mapsto [b]$.

Every cycle b of the intersection has fillings u in M_1 and v in M_2, since both complexes are acyclic; then $u + v$ proves surjectivity. If $b = \partial a$ in the intersection, the chains $u + a, v + a$ are cycles in M_1, M_2 and hence boundaries. Their sum is z, proving injectivity. For $t = -1$, the left side is zero since the acyclic complexes M_1, M_2 have vertices; the right side is in degree -2. In lower degrees both sides vanish. □

The folds used in the graph reductions preserve these homology groups.

Lemma 6.3. *A fold $v \to w$ induces isomorphisms $\widetilde{H}_t(\mathrm{Cl}(H - v)) \cong \widetilde{H}_t(\mathrm{Cl}(H))$ for every integer t.*

Proof. Pair every face σ containing v but not w with $\tau = \sigma \cup \{w\}$. The domination condition guarantees that τ is a face. Remove these pairs in decreasing order of $|\sigma|$. When a pair is removed, τ is the only proper coface of σ still present: every larger coface was removed in an earlier pair. The remaining faces form a subcomplex.

To see that removing one such pair preserves augmented homology, let K' be the resulting subcomplex of the current complex K and put $r = \dim \sigma$. A cycle of degree $r + 1$ has coefficient zero at τ, since its boundary has coefficient zero at σ. For a cycle of degree r, adding its coefficient at σ times $\partial\tau$ removes σ without changing its homology class. In other degrees no basis element is removed. Thus every class has a representative in K'.

Conversely, suppose a cycle z' in K' bounds a chain y in K. If y has degree $r + 1$, its coefficient at τ is zero, because the coefficient of σ in z' is zero. If y has degree r, add its coefficient at σ times $\partial\tau$;

this removes σ and preserves its boundary. The resulting filling lies in K'. This proves injectivity as well, including in degree -1. After all pairs are removed the complex is exactly $\mathrm{Cl}(H - v)$. □

The recursive neighbourhood condition therefore gives a homological obstruction at every level.

Lemma 6.4. *If $H \in \mathcal{D}_j$, $j \geq -1$, then $\mathrm{Cl}(H)$ is acyclic.*

Proof. Cones are covered by Lemma 6.1. Induct on j and, at a fixed level, on the number of vertices. The one-vertex graph is a cone. For a valid first deletion v, the two induction hypotheses give acyclicity of $\mathrm{Cl}(H - v)$ and $\mathrm{Cl}(L_H(v))$. The exact decomposition

$$\mathrm{Cl}(H) = \mathrm{Cl}(H - v) \cup \big(v * \mathrm{Cl}(L_H(v))\big)$$

has acyclic pieces and intersection $\mathrm{Cl}(L_H(v))$. Lemma 6.2 proves the assertion. □

6.2. **The extension and its kernel.** In a construction with two auxiliary types joined by an edge, the profiles determine which seed edges contribute to the homological obstructions. Let A be a non-empty graph, and choose $S, T \subseteq V(A)$. Form B by adding vertices p, x, y. The centre p is adjacent to all of A and to neither auxiliary vertex. The neighbours of x in A are exactly S, those of y are exactly T, and xy is an edge. The induced graph on A remains unchanged.

Lemma 6.5. *If every non-central neighbourhood of B has non-trivial reduced homology, then $S \cup T = V(A)$. The same conclusion holds if $q(L_B(v)) \neq 0$ for every $v \neq p$.*

Proof. If $a \notin S \cup T$, then its neighbours in B are $N_A(a)$ and p. The centre is an apex of $L_B(a)$. This cone has zero reduced homology and $q = 0$, contradicting either condition. □

Henceforth assume $S \cup T = V(A)$, and put

$$S_0 = S \setminus T, \qquad T_0 = T \setminus S, \qquad Z = S \cap T.$$

Define Q by deleting from A all its edges between S_0 and T_0, and no others. We call Q the *kernel of these profile data*. This name describes an operation on edges; it does not impose a homological property on Q.

Set

$$K = \mathrm{Cl}(A), \quad X = \mathrm{Cl}(A[S]), \quad Y = \mathrm{Cl}(A[T]), \quad W = \mathrm{Cl}(A[Z]), \quad U = \mathrm{Cl}(Q).$$

Every clique of Q is contained in S or in T, since it cannot meet both S_0 and T_0. Conversely every clique contained in S or in T survives the edge deletions. Therefore

$$U = X \cup Y, \qquad W = X \cap Y. \tag{9}$$

In particular this union is a clique complex, even if it is smaller than K.

6.3. **The two auxiliary stars.** To compare the seed, its extension and the remainder, we group the faces containing auxiliary vertices. For a vertex z outside a complex J, $z * J$ denotes its cone. Define the subcomplex

$$V = (x * X) \cup \big(y * (Y \cup (x * W))\big). \tag{10}$$

It consists of all faces using an auxiliary vertex together with their subfaces. Its two displayed pieces are cones and their intersection is $x * W$, also a cone. This intersection contains the vertex x,

even when $W = \{\varnothing\}$. By Lemma 6.2, V is acyclic. The faces of V on the original vertices are exactly $X \cup Y$. Consequently

$$\mathrm{Cl}(B) = (p * K) \cup V, \qquad \mathrm{Cl}(B - p) = K \cup V, \qquad (p * K) \cap V = K \cap V = U. \tag{11}$$

These are equalities of complexes.

Proposition 6.6. *For every integer* t,

$$\widetilde{H}_t(\mathrm{Cl}(B)) \cong \widetilde{H}_{t-1}(U).$$

If K *is acyclic, then also*

$$\widetilde{H}_t(\mathrm{Cl}(B - p)) \cong \widetilde{H}_{t-1}(U).$$

Under this extra hypothesis, $\mathrm{Cl}(B - p)$ *is acyclic if and only if* U *is acyclic.*

Proof. The two pieces in the first decomposition of (11) are acyclic and intersect in U. Apply Lemma 6.2. For the second decomposition the same argument applies when K is acyclic. The shift covers every possible non-zero degree; hence the last equivalence follows. □

For a seed $A \in \mathcal{D}_s$, membership of $B - p$ in a dismantling class therefore implies acyclicity of the kernel. The converse statement about membership does not follow: the proposition characterises acyclicity of the remainder, not its dismantling level.

6.4. **Original and auxiliary neighbourhoods.** The global decomposition also applies within each neighbourhood. For an original vertex, its intersection is the neighbourhood complex in the kernel.

Theorem 6.7. *For every* $a \in V(A)$ *and every integer* t,

$$\widetilde{H}_t(\mathrm{Cl}(L_B(a))) \cong \widetilde{H}_{t-1}(\mathrm{Cl}(L_Q(a))).$$

Proof. Put $K_a = \mathrm{Cl}(A[N_A(a)])$. If $a \in S_0$, its neighbourhood in B contains p, x but not y, and

$$\mathrm{Cl}(L_B(a)) = (p * K_a) \cup (x * X_a), \qquad X_a = \mathrm{Cl}(A[N_A(a) \cap S]).$$

Both pieces are cones and their intersection is X_a. The edge deletions defining Q remove exactly the neighbours of a in T_0. Edges among its remaining neighbours in S are unchanged. Thus $X_a = \mathrm{Cl}(L_Q(a))$, and Lemma 6.2 proves the assertion. The case $a \in T_0$ is obtained by exchanging the two profiles and auxiliaries.

If $a \in Z$, none of the edges incident to a is deleted. The neighbourhood contains both auxiliaries. Restrict the construction (10) to $N_A(a)$, writing

$$X_a = \mathrm{Cl}(A[N_A(a) \cap S]), \quad Y_a = \mathrm{Cl}(A[N_A(a) \cap T]), \quad W_a = X_a \cap Y_a.$$

The resulting complex V_a is acyclic by the same union-of-cones argument. Now $\mathrm{Cl}(L_B(a)) = (p * K_a) \cup V_a$, with intersection $X_a \cup Y_a$. The proof of (9), restricted to $N_A(a)$, identifies this intersection with $\mathrm{Cl}(L_Q(a))$. Lemma 6.2 again applies. Augmented chains cover all possible empty neighbourhoods. □

The auxiliary neighbourhoods have the exact descriptions

$$\mathrm{Cl}(L_B(x)) = X \cup (y * W), \qquad \mathrm{Cl}(L_B(y)) = Y \cup (x * W). \tag{12}$$

They do not contain p.

Lemma 6.8. *If $U = X \cup Y$ is acyclic, the inclusions induce isomorphisms*

$$\widetilde{H}_t(W) \longrightarrow \widetilde{H}_t(X) \oplus \widetilde{H}_t(Y)$$

for all integers t.

Proof. For surjectivity, let a, b be cycles in X, Y. Since $a + b$ is a cycle in U, it bounds a chain $w = w_X + w_Y$ there. Then $z = a + \partial w_X = b + \partial w_Y$ is a cycle in W representing the prescribed pair of homology classes. For injectivity, suppose a cycle z in W bounds chains u in X and v in Y. The cycle $u + v$ bounds $w = w_X + w_Y$ in U. The chain $u + \partial w_X = v + \partial w_Y$ lies in W and has boundary z. These arguments apply in degree -1 as well, using the augmentation, and lower degrees are zero. □

When the kernel is acyclic, the auxiliary neighbourhoods are controlled by the opposite profile subgraphs.

Theorem 6.9. *If U is acyclic, then for every integer t,*

$$\widetilde{H}_t(\mathrm{Cl}(L_B(x))) \cong \widetilde{H}_{t-1}(Y), \qquad \widetilde{H}_t(\mathrm{Cl}(L_B(y))) \cong \widetilde{H}_{t-1}(X).$$

Proof. By Lemma 6.8, the map $\widetilde{H}_r(W) \to \widetilde{H}_r(X)$ is surjective and its kernel is isomorphic to $\widetilde{H}_r(Y)$: under the lemma's isomorphism it is the projection onto the first summand.

For $t \geq 0$, write a cycle of $X \cup (y * W)$ as $u + y * w$. The equations for being a cycle are $\partial w = 0$ and $\partial u = w$. Assign to it the class of w, which lies in the kernel of $\widetilde{H}_{t-1}(W) \to \widetilde{H}_{t-1}(X)$. The assignment respects boundaries: the boundary of $a + y * b$ changes w by ∂b. Every class in that kernel has a filling u in X, so the assignment is surjective.

If $w = \partial d$ in W, adding $\partial(y * d) = d + y * \partial d$ leaves the cycle $u + d$ in X. Surjectivity of $\widetilde{H}_t(W) \to \widetilde{H}_t(X)$ lets us replace this cycle, modulo a boundary in X, by a cycle in W. That cycle bounds in the cone $y * W$. The original cycle therefore bounds in $X \cup (y * W)$, proving injectivity. Its homology in degree -1 is zero because it contains y; the claimed right side then has degree -2 and is also zero. Lower degrees vanish. Use (12) and exchange X, Y for the second identity. □

6.5. **Equivalent finite conditions and their Euler form.** Here the layer construction of Section 3 has $d = 2$, $R = K_2$ and profiles S, T. We may therefore write the auxiliaries as x_i, y_i, and $|V(G_m)| = |V(A)| + 3m$. The identities just proved separate the obstructions on original vertices from those on the two auxiliary types.

Theorem 6.10. *Assume $S \cup T = V(A)$, $s \geq 0$, $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$, and $B - p \in \mathcal{D}_{s+1}$. Then the following are equivalent:*

1. *Every non-central neighbourhood $L_B(v)$, $v \neq p$, has non-trivial reduced homology.*
2. *The clique complex of the kernel Q is acyclic, every $L_Q(a)$, $a \in V(A)$, has non-trivial reduced homology, and both $A[S]$ and $A[T]$ have non-trivial reduced homology.*

Either condition implies

$$G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1} \quad (m \geq 0).$$

Proof. The seed and remainder hypotheses imply acyclicity of K and $\mathrm{Cl}(B - p)$. Proposition 6.6 then implies acyclicity of $U = \mathrm{Cl}(Q)$. Theorem 6.7 gives equivalence of the conditions on original vertices. Theorem 6.9 gives equivalence of those on the auxiliaries. Under condition (1), the data

satisfy precisely the homological criterion in Theorem 7.2, proved independently in Section 7. It applies to the displayed layered construction and proves strictness at every subsequent level. □

The remainder hypothesis is present on both sides of this equivalence. Acyclicity of the kernel is redundant under that hypothesis, but states explicitly the property needed when starting with a kernel. The equivalence does not dispense with the remainder hypothesis. Lemma 6.5 gives the covering condition on the profiles.

Extend q from graphs to finite complexes by putting $q(J) = \sum_{\sigma \in J} (-1)^{|\sigma|}$, including the empty face. This agrees with the existing notation $q(Q) = q(\mathrm{Cl}(Q))$. Inclusion–exclusion and the zero value on cones give identities without homological assumptions.

Proposition 6.11. *For the profile data above,*

$$q(B) = -q(Q), \qquad q(B - p) = q(A) - q(Q), \tag{13}$$

$$q(L_B(a)) = -q(L_Q(a)) \qquad (a \in V(A)), \tag{14}$$

$$q(L_B(x)) = q(A[S]) - q(A[Z]), \qquad q(L_B(y)) = q(A[T]) - q(A[Z]). \tag{15}$$

If $q(Q) = 0$*, the last line becomes*

$$q(L_B(x)) = -q(A[T]), \qquad q(L_B(y)) = -q(A[S]).$$

Proof. In (10), both pieces and their intersection are cones, so $q(V) = 0$. Apply inclusion–exclusion to (11) to obtain (13). The decompositions used to prove Theorem 6.7 likewise have pieces of zero q and intersection $\mathrm{Cl}(L_Q(a))$, yielding (14). Apply inclusion–exclusion to (12) to obtain (15). Finally (9) gives $q(Q) = q(A[S]) + q(A[T]) - q(A[Z])$. □

In particular, under $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$ and $B - p \in \mathcal{D}_s$, both have $q = 0$, so $q(Q) = 0$. The non-zero Euler conditions on all non-central neighbourhoods of B are then equivalent to

$$q(L_Q(a)) \neq 0 \quad \text{for all } a \in V(A), \qquad q(A[S]) \neq 0, \quad q(A[T]) \neq 0.$$

This is the kernel formulation of Theorem 3.5. The identities remain valid if the remainder is only assumed to lie in $\mathcal{D}_{s+1}$, as in its separate relaxation. The equality $q = 0$ is not used as a test for acyclicity or dismantlability.

One further necessary condition can be read from Lemma 6.8. If S, T, Z are non-empty and Q is acyclic, the numbers of connected components satisfy

$$c(A[Z]) = c(A[S]) + c(A[T]) - 1.$$

Indeed, take dimensions in degree zero and use $\dim \widetilde{H}_0 = c - 1$. This numerical equality alone does not imply acyclicity; even agreement of dimensions in all degrees would not establish that the inclusion maps in the lemma are isomorphisms.

6.6. **Completing one edge and reducing the remainder.** The case of a single added edge lets us give an explicit reduction of the remainder. Suppose Q is a graph, ab is a non-edge, and

$$A = Q + ab, \qquad S = V(Q) \setminus \{b\}, \qquad T = V(Q) \setminus \{a\}.$$

Here $S_0 = \{a\}$ and $T_0 = \{b\}$, so deleting the edges between the exclusive profile parts recovers exactly Q.

Lemma 6.12. *If* $\mathrm{Cl}(Q)$ *is acyclic, inclusion induces*

$$\widetilde{H}_t(\mathrm{Cl}(L_Q(v))) \cong \widetilde{H}_t(\mathrm{Cl}(Q - v))$$

for every vertex v *and every integer* t.

Proof. Write $M_1 = \mathrm{Cl}(Q - v)$, $M_2 = v * \mathrm{Cl}(L_Q(v))$ and $I = M_1 \cap M_2 = \mathrm{Cl}(L_Q(v))$. The union is acyclic by hypothesis, and M_2 is a cone. The same chain argument as in Lemma 6.8, now with the acyclic union $M_1 \cup M_2 = \mathrm{Cl}(Q)$, shows that $\widetilde{H}_t(I) \to \widetilde{H}_t(M_1) \oplus \widetilde{H}_t(M_2)$ is an isomorphism. The second summand is zero. □

Thus, if Q is acyclic and all its neighbourhoods have non-trivial reduced homology, then both $A[S] = Q - b$ and $A[T] = Q - a$ have non-trivial reduced homology. The remaining obligations for Theorem 6.10 are the strict level of the completed seed and the required level of $B - p$.

There is a useful sufficient condition for the latter. For disjoint graphs C, J, write $C \star J$ for their *complete join*.

Lemma 6.13. *If* $C \in \mathcal{D}_j$, $j \geq -1$, *then* $C \star J \in \mathcal{D}_j$ *for every graph* J.

Proof. For $j = -1$, an apex of C remains universal. Induct on j. Follow a level-j reduction of C. The neighbourhood of a deleted vertex in the join is the join of its current neighbourhood in C with J, and belongs to $\mathcal{D}_{j-1}$ by induction. When only one vertex of C remains, the join is a cone and reduces at level zero, hence at level j. This proves the assertion for $j \geq 0$. □

Proposition 6.14. *For the one-edge completion, put*

$$C = Q[N_Q(a) \cap N_Q(b)].$$

The remainder $B - p$ *folds onto* $C \star C_4$, *where the cycle has vertices* a, b, y, x *in this order. Consequently, if* $C \in \mathcal{D}_j$ *for some* $j \geq 0$, *then* $B - p \in \mathcal{D}_j$.

Proof. Every $z \in V(Q) \setminus \{a, b\}$ that is not adjacent to a in Q is dominated by y in $B - p$. Indeed y is adjacent to every vertex except a, and a is not in the closed neighbourhood of z. Likewise, a vertex not adjacent to b is dominated by x. Delete all vertices outside $C \cup \{a, b, x, y\}$, retaining a, b, x, y and the common neighbours. Each chosen domination inclusion remains true after restricting to the vertices still present.

In the remaining graph, the edges on $\{a, b, x, y\}$ are exactly ab, by, yx, xa, and every vertex of C is adjacent to these four vertices. Its induced graph is unchanged from Q. The remainder is therefore $C \star C_4$; when C is empty, it is just the four-cycle. If $C \in \mathcal{D}_j$, Lemma 6.13 gives $C \star C_4 \in \mathcal{D}_j$. Prepending the displayed folds is valid at level j by nesting of the classes. □

If C is a cone, the proposition applies with $j = 0$. The argument gives a sufficient certificate of the remainder. It does not assert that a failure to dismantle C at a chosen level excludes membership of $B - p$ at that level. In particular, an unsuccessful sufficient test must not replace the original remainder condition in the equivalence theorem.

6.7. **The completion underlying the eleven-vertex seed.** Define the completed seed A on $0, \ldots, 10$ by Table 12. It has thirty-three edges. The pair $\{1, 2\}$ is an edge; define $V(Q) = V(A)$ and $E(Q) = E(A) \setminus \{\{1, 2\}\}$. Then Q has thirty-two edges and eleven vertices. For the profiles $S = V(A) \setminus \{2\}$ and $T = V(A) \setminus \{1\}$, the exclusive parts are $\{1\}$ and $\{2\}$, respectively. Hence this Q is exactly the kernel of the profile data, and adding the single edge $\{1, 2\}$ to Q gives A.

TABLE 12. The complete adjacency data of the completed seed A.

v	$N_A(v)$
0	$\{3,4,6,7,8\}$
1	$\{2,4,5,6,7,9,10\}$
2	$\{1,5,6,7,8,9,10\}$
3	$\{0,6,7,8,9,10\}$
4	$\{0,1,7,8,9,10\}$
5	$\{1,2,6,8,9\}$
6	$\{0,1,2,3,5,10\}$
7	$\{0,1,2,3,4,9\}$
8	$\{0,2,3,4,5,10\}$
9	$\{1,2,3,4,5,7\}$
10	$\{1,2,3,4,6,8\}$

This is the completed seed F_1 used in the article. The reductions of Appendix A establish its strict level and dismantle $B - p$. The homology calculation gives

$$\widetilde{H}_*(\mathrm{Cl}(Q)) = 0, \qquad \dim_{\mathbb{F}_2} \widetilde{H}_1(\mathrm{Cl}(L_Q(v))) = \begin{cases} 2, & v = 10, \\ 1, & v \neq 10, \end{cases}$$

with all other reduced homology groups of these neighbourhoods zero. Moreover, $N_Q(1) \cap N_Q(2) = \{5,6,7,9,10\}$ induces the path $10,6,5,9,7$. Proposition 6.14 therefore gives $B - p \in \mathcal{D}_0$. One such reduction begins with the folds $0,3,8 \to y$ followed by $4 \to x$, which leave the join of this path with the four-cycle on $1,2,y,x$. The displayed neighbourhood groups and Lemma 6.12 give condition (2) of Theorem 6.10. With the strict level of A proved in Appendix A, that theorem provides a homological proof of the strict levels of F_k.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only kernels
```

The report reconstructs Q by removing $\{1,2\}$, restores the seed by adding that edge, computes the displayed homology groups over $\mathbb{F}_2$, and checks the common-neighbour graph. It also compares the homology of each neighbourhood in Q with the degree-shifted group in B.

7. HOMOLOGY THROUGH SUCCESSIVE LAYERS

The kernel calculations compare neighbourhoods in the first extension. We now follow those groups through every layer of the general construction, with any number of auxiliary types. As in Section 3, write $\mathcal{T} = \{0,\dots,d-1\}$ for the type set. The coefficients and augmented chain conventions are those of Section 6. The degree shift will supply the criterion used in Theorem 6.10.

7.1. A degree shift in every non-central neighbourhood.

For complexes on disjoint vertex sets, $K * J$ consists of unions of one face from each factor. If S^0 consists of two vertices without an edge, $K * S^0$ is the union of two cones on K. Lemma 6.2 gives

$$\widetilde{H}_t(K * S^0) \cong \widetilde{H}_{t-1}(K). \tag{16}$$

This includes $K = \{\varnothing\}$, for which the join is S^0. The clique complex of the complete join of two graphs is the join of their clique complexes, since a clique is a union of a clique in each graph.

Theorem 7.1. *For $m \geq 1$ and every integer t,*

$$\widetilde{H}_t\big(\mathrm{Cl}(L_{G_m}(a))\big) \cong \widetilde{H}_{t-m+1}\big(\mathrm{Cl}(L_B(a))\big), \qquad a \in V(A), \tag{17}$$

$$\widetilde{H}_t\big(\mathrm{Cl}(L_{G_m}(x_{j,i}))\big) \cong \widetilde{H}_{t-m+1}\big(\mathrm{Cl}(L_B(x_j))\big), \qquad j \in \mathcal{T}, \quad 0 \leq i < m. \tag{18}$$

Proof. First fix $a \in V(A)$ and put $\Lambda_m = L_{G_m}(a)$. Its vertices are $N_A(a)$, all centres, and all copies of the types j for which $a \in P_j$. For $m \geq 2$, delete a centre c_i from Λ_m. The folds $x_{j,h} \to x_{j,i}$, $h \neq i$, are still valid: restrict the neighbourhood inclusions in the proof of Lemma 3.2 to $N_A(a)$ and the types present. After these folds, the remaining graph is a cone with a remaining centre as apex. If there are no such auxiliary types, it is already a cone. Lemmas 6.1 and 6.3 imply that $\mathrm{Cl}(\Lambda_m - c_i)$ is acyclic.

The neighbourhood of c_i inside Λ_m is isomorphic to Λ_{m-1}, using the same layer reindexing restricted to the vertices under consideration. Hence the decomposition

$$\mathrm{Cl}(\Lambda_m) = \mathrm{Cl}(\Lambda_m - c_i) \cup \big(c_i * \mathrm{Cl}(L_{\Lambda_m}(c_i))\big)$$

has acyclic pieces and intersection isomorphic to $\mathrm{Cl}(\Lambda_{m-1})$. Lemma 6.2 shifts homology by one degree. Iteration proves (17); at $m = 1$ it is the identity. The set $N_A(a)$ may be empty, but Λ_1 still contains p.

Now fix an auxiliary vertex $x_{j,i}$. Its neighbourhood has the vertices of P_j, all copies of types in $N_R(j)$, the vertices $x_{j,h}$ for $h \neq i$, and the centres c_h for $h \neq i$. For each $u \in N_R(j)$, fold $x_{u,h}$ onto $x_{u,i}$ for $h \neq i$. The source and target have the same adjacencies to P_j and auxiliary types. The target sees all centres in this neighbourhood, whereas the source may omit c_h. These folds therefore preserve homology, and their targets remain throughout.

What remains consists of a copy of $L_B(x_j)$ on

$$P_j \sqcup \{x_{u,i} : u \in N_R(j)\}$$

and the $m-1$ pairs $\{c_h, x_{j,h}\}$, $h \neq i$. Each pair has no internal edge. Distinct pairs are completely joined, and every pair is completely joined to the displayed copy of $L_B(x_j)$. The remaining clique complex is consequently

$$\mathrm{Cl}(L_B(x_j)) * (S^0)^{*(m-1)}.$$

Apply Lemma 6.3 and then (16) $m-1$ times. If $L_B(x_j)$ is empty, both P_j and $N_R(j)$ are empty. There are then no folds, the first factor is $\{\varnothing\}$, and the same suspension identity applies. For $m = 1$ there are no pairs, so this case too is the identity. □

7.2. **The homological criterion.** Non-zero groups in the first layer remain non-zero after every shift. Together with a reduction of the remainder, they give the following criterion for strictness.

Theorem 7.2. *Let $s \geq 0$. Suppose that*

1. *$A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$;*
2. *$B - p \in \mathcal{D}_{s+1}$;*
3. *for every $v \in V(B) \setminus \{p\}$, some integer t satisfies $\widetilde{H}_t(\mathrm{Cl}(L_B(v))) \neq 0$.*

Then for every $m \geq 0$,

$$G_m \in \mathcal{D}_{s+m} \setminus \mathcal{D}_{s+m-1}, \qquad |V(G_m)| = |V(A)| + (d+1)m.$$

Proof. Theorem 7.1 propagates non-zero reduced homology to every non-central neighbourhood in every G_m, $m \geq 1$. By Lemma 6.4, none of these neighbourhoods belongs to any class $\mathcal{D}_j$.

The case $m = 0$ is the first hypothesis. For $m = 1$, the centre has neighbourhood $A \in \mathcal{D}_s$ and its deletion leaves $B - p \in \mathcal{D}_{s+1}$. Equation (1) yields $B \in \mathcal{D}_{s+1}$. A first deletion at level s is impossible: the centre's neighbourhood is not in $\mathcal{D}_{s-1}$, and neither is any non-central neighbourhood. Since B has more than one vertex, (1) gives $B \notin \mathcal{D}_s$. This includes $s = 0$, when $\mathcal{D}_{s-1}$ is the class of cones.

For $m \geq 2$, suppose G_{m-1} has strict level $s + m - 1$. Each centre has neighbourhood isomorphic to G_{m-1}; deleting it leaves a graph in $\mathcal{D}_0 \subseteq \mathcal{D}_{s+m}$ by Lemmas 3.1 and 3.2. Thus $G_m \in \mathcal{D}_{s+m}$. A first deletion at level $s + m - 1$ would require a neighbourhood in $\mathcal{D}_{s+m-2}$. Centres fail this condition by the induction hypothesis, and non-central vertices fail it by homology. This proves strictness. The order follows from (2). □

7.3. **Comparison with Euler characteristic.** The obstruction q of Section 3 is the alternating sum of reduced Betti numbers, with a contribution from degree -1. Rank–nullity in the augmented chain complex gives

$$q(H) = \sum_{r \geq 0} (-1)^r \dim_{\mathbb{F}_2} \widetilde{H}_{r-1}(\mathrm{Cl}(H)). \tag{19}$$

Indeed, if $f_t = \dim \widetilde{C}_t$, $b_t = \dim \widetilde{H}_t$, and $\rho_t = \operatorname{rank} \partial_t$, then $f_t = b_t + \rho_t + \rho_{t+1}$. In the finite alternating sum the ranks cancel, including the term of degree -1. In particular, $q(H) \neq 0$ implies non-trivial reduced homology.

The converse need not hold. The graph $C_4 \sqcup K_1$ has five vertices, four edges and no triangles, so $q = 1 - 5 + 4 = 0$. Its two components give $\dim \widetilde{H}_0 = 1$. The boundary matrix of the four cycle edges has rank three, and there are no two-dimensional faces, giving $\dim \widetilde{H}_1 = 1$. These contributions cancel in (19). This example separates the two obstructions on a graph; it does not by itself supply initial data satisfying all three hypotheses of Theorem 7.2.

The degree shifts in Theorem 7.1 consequently give, for $m \geq 1$,

$$\begin{aligned} q(L_{G_m}(a)) &= (-1)^{m-1} q(L_B(a)), \quad a \in V(A), \\ q(L_{G_m}(x_{j,i})) &= (-1)^{m-1} q(L_B(x_j)), \quad j \in \mathcal{T}, \quad 0 \leq i < m. \end{aligned} \tag{20}$$

Indeed, shifting every homology degree by $m - 1$ multiplies the alternating sum in (19) by $(-1)^{m-1}$.

The Euler criterion in Theorem 3.5 uses $q(L_B(v)) \neq 0$ for every non-central v, together with $A \in \mathcal{D}_s \setminus \mathcal{D}_{s-1}$ and the stronger remainder condition $B - p \in \mathcal{D}_s$. Equation (19) and Lemma 2.3 imply the hypotheses of Theorem 7.2. Allowing the remainder in $\mathcal{D}_{s+1}$ is a separate change from replacing Euler characteristic by homology: the first-deletion argument at $m = 1$ also permits this weaker remainder in the Euler proof. Both criteria retain a distinct requirement on the remainder; non-trivial homology of the other neighbourhoods alone does not ensure the positive reduction.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only homology
```

The calculation checks the empty graph, a cone, $C_4 \sqcup K_1$, and an auxiliary with an empty first-layer neighbourhood. In the last example, after m layers its only non-zero reduced Betti number is $\widetilde{\beta}_{m-2} = 1$, for the tested values $m = 1, 2, 3$. The full command `python reproduce.py -extended` additionally checks all labelled seeds on one, two or three vertices, all profiles for one

or two types, and all type graphs, at $m = 0, 1, 2, 3$. Homology is computed for the subcollection with seed order at most two. These are finite checks of the identities; Theorem 7.1 proves them for arbitrary seeds and all layers.

8. The finite verification of the minimum at level two

The minimum at strict level one is $m_1 = 8$ [2]. This section explains the verification that $m_2 = 12$. The calculations exclude the finite set of eleven-vertex extensions left by the mathematical reduction below. The twelve-vertex graph H_2 from Section 4 supplies the matching example.

8.1. **Why these extensions suffice.** First let $k \geq 2$ and choose a graph J of strict level k and order m_k. If v is the first vertex of a level-k reduction, then $J - v \in \mathcal{D}_{k-1}$ by minimality. The graph $X = L_J(v)$ lies in $\mathcal{D}_{k-1} \setminus \mathcal{D}_{k-2}$: otherwise that first deletion would put J in $\mathcal{D}_{k-1}$. Since J is not a cone, v has a non-neighbour. Counting X, v and that non-neighbour gives

$$m_k \geq m_{k-1} + 2 \qquad (k \geq 2).$$

If a graph of strict level two had at most eleven vertices, we could choose J above with $k = 2$ and $|V(J)| \leq 11$. Put $U = V(J) \setminus N_J[v]$. Since $m_1 = 8$ and U is non-empty, the only possible sizes are

$$(|V(X)|, |U|) = (8, 1),\ (9, 1),\ (8, 2).$$

We use the following small-order classification [2]: every non-empty graph of order at most eight whose clique complex is acyclic over $\mathbb{F}_2$ belongs to $\mathcal{D}_1$; exactly seven isomorphism classes among these graphs lie outside $\mathcal{D}_0$, all of order eight.

If $U = \{u\}$, put $Z = N_J(u) \subseteq V(X)$. Fold each vertex of $V(X) \setminus Z$ onto v. The remainder is the complete join $X[Z] \star \overline{K_2}$, where the two new vertices are u, v. By Lemmas 6.4 and 6.3, its clique complex is acyclic. The suspension identity (16) then gives acyclicity of $\mathrm{Cl}(X[Z])$, including degree -1; in particular Z is non-empty. If $Z = V(X)$, we already have $X[Z] \in \mathcal{D}_1$. Otherwise $|Z| \leq 8$, so the small-order classification gives the same conclusion. Lemma 6.13 now puts the remainder in $\mathcal{D}_1$, and prepending the folds puts J in $\mathcal{D}_1$, a contradiction. Both cases with $|U| = 1$ are therefore excluded without enumerating nine-vertex graphs.

Only $(|V(X)|, |U|) = (8, 2)$ remains. Since $X \in \mathcal{D}_1 \setminus \mathcal{D}_0$, it is one of the seven graphs in the classification. This is why the following seven seeds, rather than a catalogue of all eleven-vertex graphs, suffice.

8.2. **Seven seeds with explicit reductions.** Use labels $0, \ldots, 7$ and specify each seed by its missing edges in Table 13. The notation ab means the unordered pair $\{a, b\}$. None has a dominated vertex.

Table 13. The seven eight-vertex seeds, specified by their missing edges.

Graph	Missing edges
X_1	01, 02, 07, 13, 16, 25, 26, 35, 45, 46, 47
X_2	02, 03, 05, 13, 14, 24, 26, 34, 57, 67
X_3	01, 04, 05, 17, 24, 26, 35, 36, 37, 56
X_4	01, 04, 05, 17, 24, 26, 27, 35, 36, 56
X_5	02, 03, 04, 12, 15, 16, 35, 37, 46, 67
X_6	01, 02, 12, 15, 26, 34, 37, 46, 57
X_7	02, 03, 05, 12, 14, 16, 34, 57, 67

In Table 14, the first vertex has a four-vertex path as its neighbourhood, so it may be deleted at level one. The subsequent folds leave the indicated cone. Each seed therefore has strict level one. Their edge counts are $17, 18, 18, 18, 18, 19, 19$. For $X_2, \dots, X_5$, the vertices of degree four induce

TABLE 14. First deletions and folds of the seven seeds. Paths list the neighbours in path order; folds take place after the first deletion.

Graph	First vertex	Path	Subsequent folds	Apex
X_1	0	$4, 3, 6, 5$	$4 \to 2$	7
X_2	3	$7, 2, 5, 6$	$2 \to 1,\ 7 \to 0$	6
X_3	0	$3, 2, 7, 6$	$3 \to 1,\ 2 \to 5$	4
X_4	0	$2, 3, 7, 6$	$2 \to 1$	4
X_5	0	$1, 7, 5, 6$	$1 \to 4$	2
X_6	1	$3, 6, 7, 4$	$3 \to 5,\ 2 \to 4$	0
X_7	0	$1, 7, 4, 6$	$1 \to 3$	2

graphs with degree sequences $(1, 1, 1, 1)$, $(0, 1, 1, 2)$, $(1, 1, 2, 2)$ and $(2, 2, 2, 2)$, respectively. The two degree-four vertices are non-adjacent in X_6 and adjacent in X_7. These invariants distinguish all seven classes.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only small-orders
```

The calculation verifies these reductions and invariants. It also reads all 13 599 graphs in the supplied catalogues of orders zero through eight, computes augmented reduced homology over $\mathbb{F}_2$, and tests membership in $\mathcal{D}_0$ and $\mathcal{D}_1$. It finds exactly the seven classes above and records an explicit isomorphism for each. The empty graph is included and has $\widetilde{\beta}_{-1} = 1$; it is not counted as acyclic. The catalogue sources and file identities are recorded in `THIRD_PARTY.md` and `data/catalogues/provenance.json`. The main command `python reproduce.py` also checks completeness independently: it tests pairwise non-isomorphism and compares the number of entries at each order with Burnside's count of the orbits of edge sets under all vertex permutations.

8.3. **All choices of the two external neighbourhoods.** For each X_i, add v, u_1, u_2. The neighbours of v are precisely $V(X_i)$; the vertices u_1, u_2 may have arbitrary neighbourhoods $Z_1, Z_2 \subseteq V(X_i)$, and their mutual edge may be present or absent. The program considers all $2 \cdot 256^2 = 131\,072$ ordered choices for each seed. It also records those with $Z_1 \le Z_2$, ordering subsets by their binary encodings. Exchanging u_1, u_2 identifies the two orders, leaving $2\binom{257}{2} = 65\,792$ choices per seed and 460 544 in total. These are choices of extension data; different choices can produce isomorphic graphs.

For each extension, a non-zero q excludes every dismantling class by Lemma 3.4. If $q = 0$, the program applies the exact recursion (1) at levels zero, one and two. Every possible first deletion is considered when necessary, and each recursive call has fewer vertices. A zero q alone never establishes membership.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only order-eleven
```

TABLE 15. The 65 792 extension choices per seed, after exchanging the two external vertices. The four columns partition the choices; the last column has $q = 0$ and lies outside $\mathcal{D}_2$.

Graph	$q \neq 0$	$\mathcal{D}_0$	$\mathcal{D}_1 \setminus \mathcal{D}_0$	$q = 0,\ \notin \mathcal{D}_2$
X_1	32823	27320	3610	2039
X_2	36048	25593	1821	2330
X_3	36283	25130	1920	2459
X_4	35171	26246	2219	2156
X_5	32323	28668	2930	1871
X_6	36573	25833	1331	2055
X_7	33543	28579	1733	1937

The program evaluates all 917 504 ordered choices and compares the classification of all 460 544 choices up to exchange with a separate implementation using sets of vertices and clique enumeration. The report gives the ordered counts, the counts in Table 15, and the counts with $Z_1 = Z_2$. For each category it checks $N_{\text{ordered}} = 2N_{\text{unordered}} - N_{\text{equal}}$. No extension has strict level two. Combined with the preceding case reduction, this proves that no graph of order at most eleven has strict level two.

The seven-root certificate in Section 4 proves that H_2 has strict level two and twelve vertices. The command below is an additional check of the article's direct verification, not a further requirement for this proof. It uses the automorphism $(4\,5)(6\,7)(8\,9)(10\,11)(12\,13)$ of H_2 to group its neighbourhoods. Four representative neighbourhoods fold to the octahedral graph $K_{2,2,2}$, whose non-zero q excludes ordinary dismantling; the other representatives have no dominated vertex. The calculation checks these obstructions and the positive reduction.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only h2
```

It verifies a level-two reduction and excludes a first level-one deletion at every vertex, retaining the original labels $2, \dots, 13$. Thus the upper and lower bounds give $m_2 = 12$. The descent inequality proved above therefore gives $m_k \geq 2k+8$ for $k \geq 2$. The constructed families give the upper bound $\min\{4k+4, 3k+8\}$; these inequalities do not determine the exact minimum for $k \geq 3$.

APPENDIX A. FINITE DATA FOR THE FAMILY OF ORDER $3k + 8$

This appendix collects the finite inputs used to verify F_k. The seed $A = F_1$ is given in Table 12, with eleven vertices and thirty-three edges. Its two profiles are

$$P_0 = V(A) \setminus \{2\}, \qquad P_1 = V(A) \setminus \{1\}, \qquad R = K_2. \tag{21}$$

The first extension $B = F_2$ adds p, x, y. It has fourteen vertices and sixty-five edges. For every $k \geq 1$, the layer parameter is $m = k - 1$.

A.1. **Reductions and neighbourhood obstructions.** Table 16 gives a level-one reduction of A. Each row deletes its first vertex from the current outer graph. The folds are performed only in that vertex's current neighbourhood and leave the listed survivor. Thus all ten neighbourhood tests pass at level zero, with twenty-three inner folds in total. The last outer vertex is 10. The only initial

TABLE 16. The level-one reduction of the eleven-vertex seed.

Deleted vertex	Current neighbourhood	Folds within that neighbourhood	Survivor
5	$\{1,2,6,8,9\}$	$1 \to 2,\ 6 \to 2,\ 8 \to 2,\ 2 \to 9$	9
2	$\{1,6,7,8,9,10\}$	$6 \to 1,\ 7 \to 1,\ 8 \to 10,\ 9 \to 1,\ 1 \to 10$	10
1	$\{4,6,7,9,10\}$	$6 \to 10,\ 7 \to 4,\ 9 \to 4,\ 4 \to 10$	10
4	$\{0,7,8,9,10\}$	$9 \to 7,\ 7 \to 0,\ 0 \to 8,\ 8 \to 10$	10
0	$\{3,6,7,8\}$	$6 \to 3,\ 7 \to 3,\ 3 \to 8$	8
6	$\{3,10\}$	$3 \to 10$	10
7	$\{3,9\}$	$3 \to 9$	9
8	$\{3,10\}$	$3 \to 10$	10
9	$\{3\}$	none	3
3	$\{10\}$	none	10

fold in A is $5 \to 2$; after deleting 5, the only fold is $9 \to 7$. The remaining nine-vertex graph has no dominated vertex. These exhaustive neighbourhood comparisons exclude a level-zero reduction, so $A \in \mathcal{D}_1 \setminus \mathcal{D}_0$.

Deleting p from B leaves a graph reduced by

$$\begin{aligned} &0 \to x, \quad 3 \to x, \quad 4 \to x, \quad 7 \to 9, \quad 8 \to y, \quad 9 \to 5, \\ &1 \to 6, \quad 2 \to 6, \quad 5 \to 6, \quad 6 \to 10, \quad 10 \to x, \quad x \to y. \end{aligned} \tag{22}$$

Each step satisfies the closed-neighbourhood inclusion in the current graph; the final vertex is y. This verifies $B - p \in \mathcal{D}_0$.

For $v \neq p$, put $B_v = L_B(v)$. Table 17 lists its complete clique coefficients, through cardinality four; all higher coefficients vanish. The alternating sum in each row is non-zero. Together with the two reductions above, these rows supply the three hypotheses of Theorem 3.5, with $s = 1$ and $d = 2$.

TABLE 17. Clique counts in the thirteen non-central neighbourhoods of B.

v	$\kappa_0(B_v)$	$\kappa_1(B_v)$	$\kappa_2(B_v)$	$\kappa_3(B_v)$	$\kappa_4(B_v)$	$q(B_v)$
0	1	8	21	20	5	−1
1	1	9	25	24	6	−1
2	1	9	24	21	4	−1
3	1	9	25	24	6	−1
4	1	9	25	24	6	−1
5	1	8	20	16	2	−1
6	1	9	24	21	4	−1
7	1	9	25	25	7	−1
8	1	9	24	21	4	−1
9	1	9	25	24	6	−1
10	1	9	25	23	4	−2
x	1	11	35	37	11	−1
y	1	11	35	36	10	−1

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only article
```

The calculation checks every displayed neighbourhood and fold, the complete lists of available folds in the level-zero exclusion, and all clique coefficients. It compares the coefficients with a separate enumeration of cliques. The files in `data/article_tables/` are verbatim copies of the corresponding tables or equation environments in the manuscript source, located by `data/article_correspondence.json`.

A.2. **Figure incidences and single deletions.** The diagrams retain the article's labels. The first represents only the added edges, so its omission of $\{1, 2\}$ does not remove that edge from the seed.

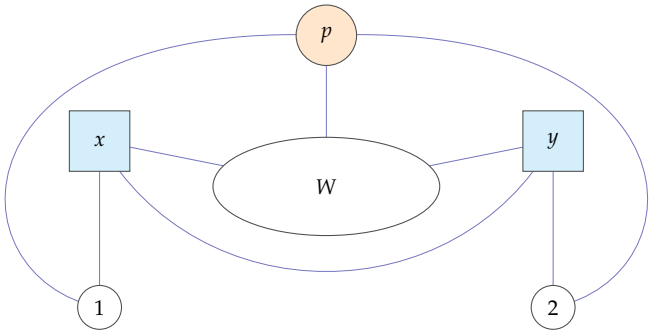


FIGURE 1. The thirty-two added edges. The ellipse W represents $V(A) \setminus \{1, 2\}$, the intersection Z in Section 6. An edge to W means adjacency to all nine vertices. Seed edges, including $\{1, 2\}$, are specified in Table 12.

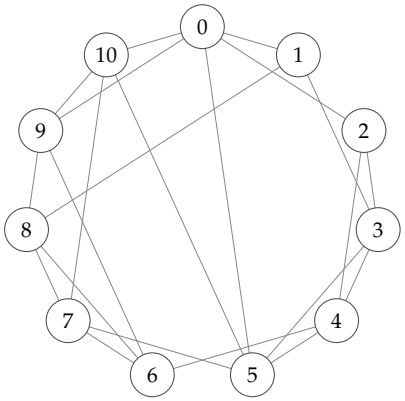


FIGURE 2. The twenty-two missing edges of the eleven-vertex seed.

The same command checks the exact edge sets: eight schematic incidences expand to thirty-two edges in the first diagram, followed by twenty-two complement edges and thirty edges in the three-layer diagram.

These facts also prove the article's single-deletion statement for $k \geq 2$. Deleting a centre from F_2 leaves a graph in $\mathcal{D}_0$ by (22); for $k \geq 3$, Lemma 3.2 gives the same conclusion. For any non-central vertex v, (20) and Table 17 give $q(L_{F_k}(v)) \neq 0$. Since $q(F_k) = 0$, Lemma 3.3 gives $q(F_k - v) =$

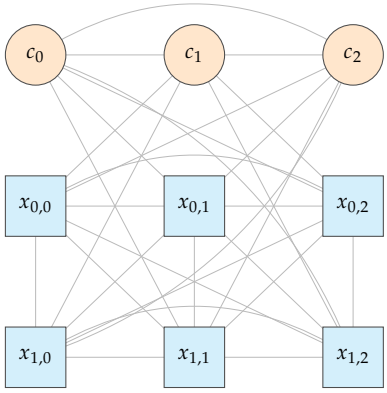


FIGURE 3. The thirty edges on the nine vertices added in three layers. Columns are layers; the seed and its incident edges are omitted.

$q(L_{F_k}(v)) \neq 0$, so Lemma 3.4 excludes every dismantling class. This statement concerns one deletion at a time.

Reproduce this calculation. From the repository directory, run

```
python reproduce.py -only deletions
```

The calculation tests every single-vertex deletion from F_2, F_3, F_4. The finite range is a check of the universal result proved in the article.

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